Karnataka Residential Educational Institutions Society

# ICT Class 8 - Practical Lab Exam 1

## Practical Lab Exam 1

Class: Class 8

Assessment Type: Practical Lab Exam 1

Total Marks: 20

Duration: 45 minutes

Topics Covered: Python Functions and Data Structures

## General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

# Lab Tasks (15 marks)

## Task 1: Function-Based Calculator (5 marks)

Write a Python program that:

- Defines functions `add(a, b)`, `subtract(a, b)`, `multiply(a, b)`, `divide(a, b)`
- Handles division by zero with proper error messaging
- Uses a `menu-driven` approach where the user selects an operation
- Calls the appropriate function based on user input
- Repeats until the user chooses to exit
- Displays results with proper formatting

## Task 2: List Operations Manager (5 marks)

Write a Python program that:

- Creates a list of at least 15 student names (user input or predefined)
- Implements a menu with options: Add, Remove, Search, Sort, Reverse, Display All
- Uses appropriate list methods for each operation
- Includes input validation (e.g., cannot remove a non-existent name)
- Displays the list after each operation

## Task 3: Dictionary-Based Marks Manager (5 marks)

Write a Python program that:

- Creates a dictionary to store marks for 8 students in subjects: Math, Science, English, SST
- Allows the user to:
  - Add a new student with marks
  - Update marks for an existing student
  - Display all students and their marks
  - Calculate and display the average per student
  - Find the topper (highest average)
- Uses nested dictionaries and proper formatting for output

# Viva Voce (5 marks)

1. What is the difference between a function definition and a function call? (1 mark)

2. Can a function have both default parameters and variable-length arguments? Give an example. (1 mark)

3. Why are dictionaries preferred over lists for storing key-value pairs? (1 mark)

4. What is the difference between `remove()` and `pop()` for lists? When would you use each? (1 mark)

5. How does Python handle memory when you create a deep copy vs a shallow copy of a nested list? (1 mark)